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The food subsystem is an integral part of ALS. Purdue and Alabama A&M Universities are continuing to address key components of food processing, food safety, energy requirements, and the effect of long term exposure of food to radiation. It is also important to consider that the research, although space based, has terrestrial benefits as well. Research to understand the affects of radiation has begun using various food stuffs. Experiments have shown that there is a dose dependant relationship in the decrease of products of primary lipid oxidation and an increase in the secondary products. Preliminary results also showed negative affects on the antioxidant capacity ascorbic acid and the wheat cultivars Apogee and Perigee while terrestrial cultivars were not affected. Packaging scenarios have also been investigate to examine potentials for reduction of waste including biodegradable packaging demonstrating areas in which significant mass could be eliminated. To ensure food safety and shelf life from microbial contamination of potential salad crops research is being performed to examine the effect of various forms of physical and chemical control strategies and combinations. Synergistic effects using the combination of chemical and physical methods showed the greatest reduction in a majority of the combinations. Research is also continuing to identify potential areas of contamination and control strategies using an integrated HACCP (hazard analysis and critical control points) expanded to evaluate the ALS system as a whole for identification of key parameters and possible points of contamination from the various waste processing regimes, hydroponics, and tilapia waste water. Phage based pathogen detection has progressed from proof in principle experiments utilizing M13 bacteriophage. Significant progress is being made on the recombinant bacteriophage for pathogen detection. Recent collaboration with industry has also provided a potential significant alternative to the integrated pathogen detection food packaging format. Bruce Applegate Focus Area Group Leader – Food Safety and Processing For Pathogen Dection in Food: The overall goal of this research is to harness the power of bacterial phage display to develop a biological amplifier for the detection of small numbers of pathogenic organisms in potable water and foods. The fundamental hypothesis of this research is: Bacteriophage can be genetically modified using recombinant DNA technology to produce an antigenic peptide which is only expressed in progeny phage after infection of a specific viable pathogenic organism. The hypothesis will be tested by addressing the following objectives:
1. Genetically modify and propagate modified bacteriophages for the detection of viable pathogens. 2. Recover and purify modified bacteriophages using affinity chromatography. 3. Develop an assay using the modified bacteriophages to detect pathogenic organisms in potable water and food using an ELISA format. 4. Integrate the developed bacteriophage assay into novel food packaging materials.

For Food Processing and Packaging: 1. Optimization of Food Quality: To optimize food quality by determining what factors will limit shelf-life, what countermeasures are available to minimize the impact of these factors, what foods and ingredients will have a shelf-life less than the duration of a mission, and how quality traits and composition of foods and crops are affected by storage conditions. 2. Integrated Package Safety Indicator System: To ensure the safety of a food prior to consumption, this collaborative effort with Dr. B. Applegate will integrate a pathogen detection system into a food package to facilitate pathogen detection. 3. Equivalent System Mass: To provide equivalent system mass parameters for food processing unit operations and packaging scenarios.

For Optimal Food Safety: The primary goal of this project is to develop a HACCP system which can be used for the validation and testing of food products for ALS mission and determine the critical points in processing and production of packaged products provided from plants produced by Purdue and AAMU researchers, and develop a HACCP and food safety system to monitor and validate the effectiveness of reducing any hazards in these packaged products. For Detection of Pathogens in Food: The research will generate and purify phages that are designed to selectively detect water and foodborne pathogens. Using specially designed markers that are inserted into the DNA of the phage, the expression of the phage can be tuned to reflect the presence (or absence) of the target cell as well as give a measure of concentration and type of cells in a given sample. Proteins displayed on the outside of the phages can be used as “handles” to distinguish one type of phage from another, much like a license plate will identify one car from another, even if the cars are alike in every way except for the owner. Since the protein is displayed on the outer surface of the phage and since it reflects a special characteristic of the DNA contained inside the phage particle, this is referred to as phage display. The proposed project combines the knowledge and experiences of fundamental molecular biology of phages and applies this knowledge to existing technologies in both the molecular construction and detection of the modified bacteriophages. The assay will provide an alternative to current culturing methodology and nucleic acid amplification technology which are time, labor, and equipment intensive. This technology platform can be integrated with antibody-based assays coupled with impedance based spectroscopy, biochips, fluorescence microscopy, and enzyme-linked immuno-assays (ELISA’s) which address the diverse earthbound needs of detection methods for food and waterborne pathogens. However these technologies can require significant equipment expense and mass. Therefore a test strip format which will provide low mass and multiplexed detection of a variety of pathogens could be developed to be utilized for direct testing of potable water and food for both coliforms and pathogens. This assay could be utilized as a stand-alone kit to test potable water or in an integrated food packaging format, which will allow the testing of foods prior to their removal from the package. The potential integration of the phage bioamplification step with ELISA’s will provide a sensitive and low mass approach to both water and food safety.

For Food Processing and Packaging: Food must be safe, nutritious, and acceptable throughout a long duration mission to maintain the health, well-being, and productivity of the astronauts. In addition to a pre-packaged food supply, the ALS NSCORT proposes to utilize crops to provide food and oxygen for astronauts. Research is required to better understand the ability to convert edible biomass into safe, nutritious, and acceptable food products in a closed system with many restrictions (mass, volume, power, crew time, etc.). An understanding of how storage conditions encountered in a long-term space mission will impact food quality is also needed. High levels of radiation encountered on a mission to Mars will provide a threat not only to human health but to the stability of the food, hence radiation is one of the top three concerns for a Mars mission (Wald, 2003). The focus of this project is to provide the highest quality food possible for the duration of a mission by combining shelf-stable extended shelf-life foods and bulk ingredients with crops grown in space, to characterize the impact of space-relevant radiation doses on food quality and antioxidant capacity, and to provide quantitative measures of ESM components related to the food system when possible.

For Optimal Food Safety: Foodborne diseases are estimated to cause ~76 million illnesses, 350,000 hospitalizations, and 5,000 deaths annually in the United States (Mead et al., 1999). Therefore, foods contaminated with pathogenic microorganisms, such as Salmonella spp., Staphyloccocus aureus, Campylobacter jejuni, E. coli O157:H7 and Listeria monocytogenes remain a major health concern for both NASA and the food industry.

Currently, several technologies and intervention steps have been evaluated for inactivation of these pathogens in salad crops (lettuce, onion, carrots, etc.), such as heat treatment, sanitizers, disinfectants, irradiation, microwave radiation and pulsed electric field to name a few. These technologies in combination with the implementation of Hazard Analysis

	Critical Control Points (HACCP) plans for NASA's Advanced Life Support (ALS) is important in reducing or eliminating potential microbial threats possibly present in hydroponically grown and packaged food products.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Further development of nano-scale detection of pathogens in packaged foods Development of novel food products and technologies for vegetarian and health-food markets
	For Pathogen Detection: Salmonella spp bacteriophage: To facilitate the construction of a modified P22 bacteriophage for Salmonella spp. it was necessary to construct a recombination vector for insertion of a modified tail spike protein. The vector was modified from a previously constructed vector pTP369 (Casjens et al). Plasmid pTP369 contains a 2558 bp region of the P22 genome corresponding to the region containing gene 23 (antitermination protein), gene 13 (lysis protein), gene 19 (lysozyme) gene 15 (lysis control), orf 201 (unknown protein) and orf 80 (unknown protein). The recombination vector was constructed by removing approximately 1 kb of the P22 genome in the region of orf 201 and orf 80 to allow the insertion of DNA for recombination. A multicloning site and a TA cloning site were inserted to facilitate rapid insertion of modified DNA to construct the appropriate epitope. Primers were utilized to amplify the tailspike protein from P22 with the appropriate His modifications. We are currently adding the appropriate promoter configurations to allow repressed expression of the His modified tailspike protein in the preparative host strain. The preparative host strain repressor gene cassette was constructed for insertion into the preparative host strain genome. Work is continuing on inserting the modified tailspike gene in the previously constructed P22 recombination vector. Appropriate promoter and terminator configurations were added to the modified tailspike protein to allow repression of expression of the His modified tailspike protein in the preparative host strain. Preparative host strain containing the lacI repressor gene cassette was constructed. E. coli O157:H7 bacteriophage: Phage based detection of E. coli O157:H7 will be accomplished using bacteriophage fV10. Bacteriophage fV10 was originally isolated by R. Khakhria and has been shown to specifically infect many strains of E.coli O157:H7. It has a genome of approximately 42 kb and is classified as a temperate phage (can form lysogens). The phage was obtained from Dr. Rafiq Ahmed at the National Laboratory for Enteric Pathogens in Winnipeg Canada. To determine the specificity of fV10, a previously characterized library from pathogenic E. coli outbreaks was screened for fV10 susceptibility. Environmental isolates were also screened to provide evaluation of false positives. The assay consists of a simple plaque assay using a previously prepared phage suspension with a titer of approximately 2 x 10 -3 plaque forming units per mL. Using this assay, 187 strains of E. coli were tested. Of these samples, 106 were known to be isolates of E. coli O157:H7 and 81 wereknown to be non-O157:H7. Results showed 93 positives and 81 negatives. The data correlated with the previous identifications. However 13 of the previously identified positives did not form plaques and were investigated further and found to be contaminants. No false positives were detected. The complete geneome of phiV10 was sequenced to provide data which will allow a more deliberate construction of the recombinant phage avoiding the use of transposon mutagenesis. Sequence analysis determined the FV10 genome to be 39,104 bp long with a G+C content of 49.0%. It contains 56 proposed open reading frames (ORFs). Functions were proposed for 19 of the 56 predicted proteins. The genomic organization and hypothetical proteomic composition of FV10 reveal a probable common ancestry with the Salmonella enterica Group E1 phage e15. Forty one hypothetical FV10 proteins have homologs in e15, thirty of which are more similar to their e15 homologs than to any other protein. The serotype conversion genes of e15 are absent from FV10. A putative acyltransferase gene occupies a position in the FV10 genome where serotype conversion genes are located in e15. A 2425 bp cluster of seven ORFs located between the putative DNA replication and terminase genes has considerable sequence identity with DNA from E. coli O157:H7phage VT2 Sa. No known bacterial virulence genes were found in the FV10 genome. The apparent host specificity of ?V10, its ability to lysogenically convert without transducing host genes, and its lack of identifiable virulence determinants suggest that ?V10 should be useful in biotechnology and that its use in biotechnology should not constitute any undue hazard, making it a good candidate for detection assays. A similar recombination system for insertion of foreign DNA into the phage genomes as described for P22 above was developed for the modification of the E. coli O157:H7 bacteriophage FV10. This was accomplished by isolating a FV10 lysogen of E. coli O157:H7. The lysogen was further characterized and shown to be inducible for the lytic phenotype. The strain was subsequently transformed with pKD46,which contains an arabinose inducible recombinase. This system allows the electroporation of linear DNA containing foreign genes flanked by DNA sequences from regions of the phage genome facilitating insertion of reporter genes into the phage genome. Initial work has identified regions for modification, allowing insertion of reporter genes which do not affect the phage life cycle. We are also utilizing pKD46 for analysis of the unknown open reading frames found in the genome of bacteriophage FV10. Key parameters were determined for optimization of the recombination system for reporter gene insertion and gene inactivation. Primers were designed to begin the systematic evaluation of the unknown open reading frames. Amplicons were constructed for the systematic disruption of identified open reading frames in phi V10. We are continuing the recombination and lysogen rescue experiments. Listeria monocytogenes bacteriophage: Initial screening of a Listeria monocytogenes library to isolate a lysogenic bacteriophage has begun. Preliminary results suggest success and are currently evaluating the isolated bacteriophage and continuing to screen the library. Sample Preparation: Another important aspect in the long term goals of this research is the application of the multiplexed phage assay in a food packaging format. We have begun to examine key parameters to allow integration of the assay into a package format in which minimal sample preparation is essential. We used a previously developed two-component system consisting of T4 bacteriophage and bioluminescent Escherichia coli lux cells to evaluate the effect of food components (ground beef) or laboratory media (0.1% peptone water and LB) on phage infectivity. E. coli lux cells serially diluted using 0.1% peptone water, LB broth, and LB with 10% (w/v) ground beef were mixed with varying concentrations (0-8.4 log pfu/ml) of T4 bacteriophags, and phage infectivity was monitored over time using a luminometer. Bioluminescence intensities from E. coli lux cells (8.7 log cfu/ml) decreased over time in the presence of increasing concentrations of T4 phages with a dynamic range of 3.4-8.4 log pfu/ml in LB broth, while peptone did not exhibit any differences of bioluminescence intensities between samples. We also found that phage infectivity occurred in LB with 10% ground beef without any significant influence of food components or natural flora in ground beef. In addition, it was confirmed that 0.5% sodium chloride supplemented in LB broth was a key component affecting phage infectivity. These initial results are being pursued to develop the assay media to be incorporated into the self contained food package assay.

Task Progress:

Packaging for pathogen detection:

As previously mentioned above the long term goal of the project is to incorporate the phage detection assay in food packaging. The initial proposed product would contain a lateral flow assay for the multiplexed determination of multiple pathogens. We have begun collaboration with Embedded Concepts, which has patented technology which allows the embedding of antibodies into plastics. This innovation would greatly enhance the potential for integration of the phage based assay into food packaging material without the need foe lateral flow simplifying development of stand alone kits involving sample preparation and detection all in one. To evaluate the embedding process we are currently exploiting an inducible recombinant M13 with a streptavidin-binding epitope which was previously developed to demonstrate the proof in principle of the assay. Phage samples were highly purified using detergents and dialyzed with water to reduce the presence of background particulates during imaging. Phage samples were pipetted onto commercially-available streptavidin-coated microscope slides and allowed to adhere at room temperature for 15 min. The sample was then pipetted away and the slide was washed with water to remove nonbinding phage. Samples were then analyzed for binding using Atomic Force Microscopy (AFM). . As the AFM images show, there is a large difference in the number of phage binding to the streptavidin-coated slide depending upon whether the phage expressed the modified copy of the surface protein or not. The wild-type negative control sample did not bind to the streptavidin-coated slide while the recombinant phage not expressing the modified copy of the protein (gIII) (uninduced) showed minimal binding compared to that of the phage expressing the modified copy. AFM was utilized to provide visualization of the binding to determine the protocols for sample preparation for the analysis of the embedded streptavidin. Our goal is to use this M13 phage system in conjunction with AFM to examine both the surface density and binding activity of plastics embedded with streptavidin to provide proof in principle experiments for the assay in this format..

For Novel Food Packaging and Processing:

Research efforts for 2004-2005 focused on: 1) characterizing the effects of space-relevant radiation doses of gamma-radiation on pure oils and antioxidants as well as wheat (crop and/or bulk ingredient) and identifying threshold radiation levels for quality changes, 2) characterizing the impact plant cultivars and growth conditions have on nutrient profiles, antioxidant capacity, functionality, and acceptability of foods (starting with sweet potato and wheat, adding strawberry, carrot, tomato, and peanut beginning Sept. 2005 in collaboration with Dr. Cary Mitchell), and 3) estimating packaging waste generated by different food storage scenarios.

Radiation Dosimetry Verification of absorbed radiation dose is an integral part of the experimental design for our studies. A Fricke dosimeter was used to verify differences in radiation treatment dose from the Gammacell 220 used to expose samples to varying levels of radiation. An aqueous ferrous sulfate solution was irradiated at different levels, and absorbance was read spectrophotometrically at 304nm. Significant differences (a=0.05) were present between all doses tested (0, 3, 10, 50, and 100Gy).

Radiation Effects on Oils Radiation can initiate a process known as autoxidation, where an initiator species reacts with the lipid, removes a hydrogen, and produces a lipid free radical. This lipid free radical can then react with other fatty acids and produce an accelerating chain reaction of similar events during propagation. Products of this oxidation of fatty acids include peroxides, alcohols, aldehydes, and carbonyls. Further reactions with these products lead to off odors and flavors accustomed to rancid fats. Additionally, essential fatty acids may be oxidized, and this could compromise the long-term health status of the astronauts depending on the magnitude of the essential fatty acid loss. Results from a study in which soybean oil was irradiated at 0, 3, 10, 100, and 1000Gy and stored at 65ºC, showed an increase in both primary and secondary products of oxidation up to 100Gy and a drop between 100Gy and 1000Gy. Soybean oil samples run concurrently but stored at 25ºC showed no significant changes in oxidation levels. Based on literature values, it is estimated that one day of storage at 65ºC equates to one month of storage at 25ºC. Current research is designed to further explore the oxidation response between 100 and 1000Gy using soybean and peanut oils irradiated at 0, 3, 10, 100, 200, 400, 600, 800, and 1000Gy stored at 65ºC, as well as to extrapolate accelerated study results to estimate shelf-life of oils following exposure to varying radiation levels. A study using menhaden oil irradiated at levels between 0 and 100 Gy followed by storage at 65ºC for up to 21 days showed a dose dependent correlation between decrease in primary products of lipid xidation (conjugated triene) and formation/increase of secondary products (TBARS).

Radiation Effects on Antioxidants Radiation doses during transit and on the surface of Mars are expected to be as high as 3 sieverts, excluding solar events. Studies have shown that antioxidants may provide long term health protection from oxidative stress caused by radiation exposure; therefore, to counteract the impact of elevated radiation exposure on astronaut health, consumption of antioxidants will be important. In addition to vitamin supplements, antioxidants within foods will also provide protection. It is important to identify how storage conditions and elevated radiation will impact the antioxidant capacity and stability in both nutritional supplements and foods. Treatment of 500?M Trolox (water soluble vitamin E analog) solutions with 0, 3 ,10, 50, or 100Gy gamma radiation dose showed significant difference (?=0.05) between the Trolox solutions irradiated at the three lowest doses (0, 3, and 10Gy) and the highest dose of 100Gy using the ferric reducing antioxidant power (FRAP) assay.Results from a study using ascorbic acid exposed to 0 or 10 Gy radiation show that the 10Gy radiation treatment decreased the power of the ascorbic acid to act as a reducing agent in the FRAP assay. Antioxidant capacity preliminary results for wheat cultivars exposed to select radiation levels show that the antioxidant capacity of Apogee and Perigee were affected by the radiation while terrestrial cultivars were not significantly impacted. This work is currently being replicated. Continued work with model systems of Trolox, ascorbic acid, and other antioxidants as well as with crops and dietary supplements is ongoing in an effort to relate decay of antioxidant power and impact of radiation dose to stability and antioxidant capacity of antioxidants provided by dietary supplementation and in foods.

Radiation Effects on Wheat Wheat is the most studied candidate crop and has the highest priority. Apogee and Perigee cultivars of wheat were developed at Utah State University to yield high amounts of wheat berries with a minimum amount of inedible crop waste. While growth conditions have been characterized, further analysis of these cultivars must be conducted to characterize the protein, lipid, and starch functionalities related to food quality, as well as antioxidant capacities, and to compare the food-functionality of these cultivars to common cultivars used in the food industry. Ilan Weiss, M.S. student who graduated in 2003, conducted initial experiments to compare Apogee and Perigee (field and hydroponically grown) to commercial wheat cultivars; however, due to limited sample size and no replicates, Adam Stoklosa, current M.S. student, is continuing this work and growing both Apogee and Perigee himself in a greenhouse at Purdue. In addition to this work, because wheat can be shipped as a bulk ingredient and/or grown in space, studies must be conducted to determine the effects of radiation exposure on the quality and functionality of

wheat.

Wheat Protein Characterization. Wheat gluten (protein) functionality can be indirectly measured using a mixograph. In a mixograph, the plasticity and mobility of dough subjected to a prolonged, relatively gentle mixing at a constant temperature is measured. The amount of force required to mix the dough indicates the dough strength and dough stability, which in turn indicates the amount of glutenin/gliadin interactions which have occurred. Following exposure to low doses of radiation (0 to 10 Gy), peak mixing times decreased and gluten interactions improved. Testing at higher doses indicated longer mixing times may be caused by protein degradation. Apogee and Perigee cultivars were found to absorb 25% more water than terrestrial cultivars and also required the most time to develop.

Preliminary results found no differences in banding patterns of wheat proteins from all cultivars used in this study on SDS-PAGE, indicating a similarity in protein composition, although relative amounts of the proteins may have varied between cultivars based on the intensity differences observed (data not shown). Further work is being done to characterize the protein structures. In an initial study investigating the impact of radiation on protein physical structure, AFM images captured differences between control and irradiated samples. The surface of gliadin in its native state is hydrophobic. When exposed to a hydrophilic surface such as the diamond used in this study, the gliadin aggregates with itself and repels from the surface. Upon exposure to radiation, increased availability of hydrophilic amino acids occurs (Matloubi 2004). This apparently occurs after gliadin has been exposed to 10 kGy radiation, and the gliadin protein is attracted instead of repulsed by the diamond surface, and a more compact adhesion occurs.

Wheat Starch Characterization The starch content of wheat appears to be inversely related to the protein content of the cultivar. Most wheat cultivars fall into the range of 63-72% with hard and soft wheat having a starch content of approximately 64% and 69% respectively. Starch acts as a temperature activated water sink in baked cereal products. As temperature increases and starch gelatinizes, the starch competes with other components for the available water in the system. Therefore, starch often determines the structure of the baked product. The extent of starch pasting is a function of the availability of water to the starch granules. The final state of the starch contributes to the textural attributes of the baked product. Both starch concentration and structure contribute to starch functionality. Results from a study that exposed wheat to various radiation doses indicate that starch granules from the Pergiee cultivar are the most susceptible to mechanical and radiation damage. The other cultivars, including Apogee, appear to resist significant damage.

Wheat Lipid Characterization Wheat contains 1.5-3% lipid depending on the cultivar and growing conditions. The majority of lipids in wheat are contained within the aleurone layer of the endosperm and the germ. Linoleic acid, oleic acid, and linolenic acid constitute the majority of fatty acids found in wheat. These lipids provide numerous health benefits and linoleic and linolenic are essential fatty acids, but due to their number of unsaturated bonds, they also are susceptible to oxidative rancidity. The products of oxidative rancidity reactions have a significant impact on flavor stability and texture during food processing and can impact overall product acceptability. In a preliminary comparison of lipid oxidation between wheat cultivars measured using the TBARS assay , the Parshall cultivar does have elevated levels of oxidation.

Food Packaging Waste Scenarios The goal of an Advanced Life Support (ALS) system on a planetary outpost is to be bioregenerative. The packaging materials currently used will accumulate as unusable waste. Because handling unusable waste in an enclosed, bioregenerative life support environment will be difficult, it is imperative to investigate alternate packaging scenarios that will both minimize waste and maximize product quality (barrier properties are essential). Using the BVAD waste assumptions, package waste mass for the 1080 day mission to Mars would be between 1555 kg and 1750 kg. All comparisons of other packaging options obtained from a survey of literature/commercial products were made to reference values from Levri et al. (2001). Based on these estimates, spreadsheets were developed to predict packaging waste generated for a variety of food storage and consumption scenarios, and this data was presented at the 2005 Institute of Food Technologists' Annual Meeting. To summarize conclusions, removing the rehydratable foods following the 240 day outbound flight has the greatest single impact on package waste, reducing the waste by ~17% if irradiated, IMF, and/or thermostabilized foods are substituted. Removing 90% of single-serve beverages during the 600 day surface stay reduced package waste by ~11% Biodegradable packaging may not be a viable option for space foods due to barrier property limitations and strict control of food quality and safety. Impact of biodegradable packages on solid waste handling systems is unknown. Depending on amount of mass reduction over current package formats, lighter-weight high-barrier package materials that meet NASA constraints could reduce package waste (with an average 1 gram per package reduction, the total package waste would be reduced by ~8%).

For Optimal Food Safety:

Sampling plans for both whole tomatoes has been completed. Currently, experiments are underway to detect E. coli, Salmonella, and selected coliforms from hydroponically and conventionally grown salad crops, including mushrooms. The efficacy of selected sanitizers and combination with pulsed light sterilization as a corrective action for reducing or eliminating pathogenic microorganisms on surface of whole tomatoes is being investigated. Also, preliminary studies were conducted to determine the recovery of selected mesophiles and psychrophiles (Citrobacter, Pseudomonads, E. coli, and coliforms) from a hydroponic system growing salad crops.

A. Inactivation of Salmonella by pulsed light sterilization and sanitizers

To determine the role of sanitizing agents and pulsed light sterilization on reduction of a high inoculum of Salmonella spp. on the surface of tomatoes, we selected several sanitizers, based on a previous study, which was shown to be highly effective in reducing Salmonella on the surface of tomatoes. Three sanitizers, peracetic acid, hydrogen peroxide and Prosan®, a commercially available product were all chosen based on a previous studies (English and Williams, 2004) and its ability to leave behind limited amounts of chemical residues. Two stain mixtures of Salmonella was obtained from the AAMU Food Microbiology Laboratory Culture Collection to determine the efficacy of sanitizers and pulsed light sterilization on whole tomatoes stored for 8 days at 25°C. Cells were grown in 150 ml of tryptic soy broth (Difco, Decton Dickson, Sparks, Md.) at 37°C for 24 h, and harvested by centrifugation at 3,300 x g for 25 min at 4°C.

Pulsed UV-light treatment. Pulsed UV-light treatment was conducted with a laboratory scale, batch-pulsed light sterilization system (SteriPulse-XL 3000; Xenon Corporation). This unit generated approximately 5.6 J/cm2 per pulse on the strobe surface, for an input voltage of 3,800 V and with three pulses per second, as per manufacturer's instructions. The output of the UV-pulsed light sterilizer followed a sinusoidal wave pattern, with 5.6 J/cm2 per pulse being the peak value of the pulse. The power values of the treatments were based on the peak value. The duration (pulse width) was 360 µs. The inoculated whole tomatoes treated with various sanitizers were treated under pulsed UV light for 60 s at a distance of 2.5 cm from the UV strobe. Similarly, untreated (no sanitizer treatment) were treated under pulsed UV light

for 60 s at a distance of 2.5 cm from the UV strobe. After pulsed UV-light sterilization, tomatoes were stored for a period of 8 days at 25°C in a temperature controlled incubator. At day 0, 2 and 8 days, samples were randomly selected from each treatment group and microbiological analysis was conducted.

Microbial analysis: For inoculated tomatoes, untreated samples and samples immediately after pulsed UV-light treatment (day 0) were analyzed for surviving populations of Salmonella. Briefly, a 1 ml sample from tomato rinse solution was serially diluted with 0.1 M phosphate buffer. This was followed by spiral plating of a 0.1 ml sample onto plate count agar. After incubation at 37 C for 24h, the colonies were enumerated and log reduction was calculated by subtracting the log value of control from that of treated sample. Each experiment was repeated four times.

Data can be summarized as follows: To demonstrate the effectiveness of pulsed light and sanitizer treatment on surface of whole tomatoes, Salmonella cells were inoculated onto the surface of whole tomatoes, treated with selected sanitizers and pulsed UV-light up to 60 s at a distance 2.5 cm from the UV strobe. The corresponding power of pulsed UV light at the surface was 1,008 J/cm2/s for 60 s treatment time. The viability of Salmonella on the surface of whole tomatoes was significantly reduced in both 1% hydrogen peroxide and peracetic acid treatment when used in combination with pulsed UV-light sterilization. At day 0, approximately log 6 cfu/tomato of Salmonella was recovered from both water and no treatment group, respectively. However, 1% prosan in combination with pulsed light sterilization resulted in approximately log 5 cfu/tomato of Salmonella survival. At both 2 and 8 days of storage 1% peracetic acid with pulsed UV-light sterilization was the most effective in reducing the viability of Salmonella on the surface of the tomatoes . A complete inactivation of Salmonella on tomatoes dipped in peracetic acid (1%) and treated with pulsed UV-light, resulted in a 6 log reduction . Our results indicated that 1% hydrogen peroxide in combination with pulsed light treatment resulted in a significant (p=0.05) reduction of Salmonella on the surface of the tomatoes when compared to water (control) and pulsed light treatment used alone, that the least effective treatment was 1% prosan and pulsed UV-light treatment at both 0 and 8 days of storage. Nevertheless, very few studies have examined the microbial communities in hydroponic water distribution systems and their trophic relationships. A preliminary study was conducted to examine the microbial ecology water in a hydroponics distribution system and it role in salad crops. In this study, our hypothesis was that most plant or waterborne microorganisms can thrive in a hydroponic system during growth of salad crops, due to the abundant of nutrients present in the water and generally created by the plants. The objectives of this study was to determine the microbial ecology of mesophilic and psychrophilic microorganisms in a hydroponic distribution system growing salad crops (lettuce, sweet potatoes, beets and carrots). Enumeration of bacterial populations included mesophilic aerobic bacteria, psychrotrophic aerobic bacteria, coliforms, generic E. coli and pseudomonads. For determination of total plate counts (TPC), appropriate dilutions were plated in duplicates on plate count agar. For enumeration of total coliforms and generic E. coli populations, 3M petrifilm coliform/E. coli Count Plates and Violet Red Bile Agar was used. For enumeration of citrate producing bacteria and Pseudomonads, Citrobacter isolation agar and Pseudomonas isolation agar were used, respectively. All plates were incubated at 37°C and enumerated according to the manufacturer's directions, except for the psychrotrophic aerobic plates that were incubated at 20°C. The mesophilic aerobic TVC and coliform counts were determine after 24 h of incubation, the E. coli counts taken after 48 h, and the psychrotrophic TPC counts were taken at 3 days. Figure 1 and 2 depict the microbial profiles of hydroponic water distribution system over a 12-week period. Figure 1 represent total Pseudomonads, Citrobacter and generic E. coli detected. There was a significant decline (p = 0.05) in the numbers of mesophilic microorganisms in the hydroponic system over 12 weeks of of the experiment.. The reduction was sustained after 7 weeks. After seven weeks, Pseudomonads were the only bacteria detected by traditional plating methods. Figure 2 illustrates the psychotropic and total plate counts from hydroponic water. The total viable counts in the hydroponic system did not significantly (p=0.05) decline until week 11. The number of psychotropic microorganisms followed a similar trend to the Citrobacter and E. coli enumerated under mesophilic conditions.

Bibliography Type:	Description: (Last Updated: 04/16/2020)
Articles in Peer-reviewed Journals	Perry, L., Shroyer; M., Minocha; U., Farris, L.; and Applegate, B. "Genomic analysis of Escherichia coli O157:H7 bacteriophage FV10 " Journal of Bacteriology. In review. , Jun-2005
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